

CS:4980

Foundations of Embedded Systems

Asynchronous Model

Part II

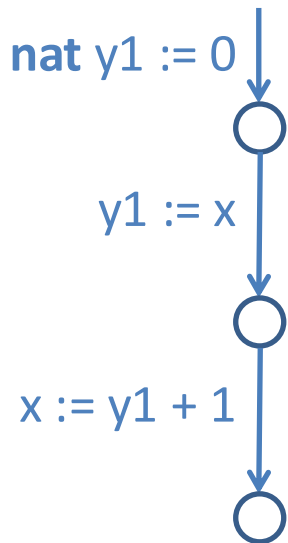
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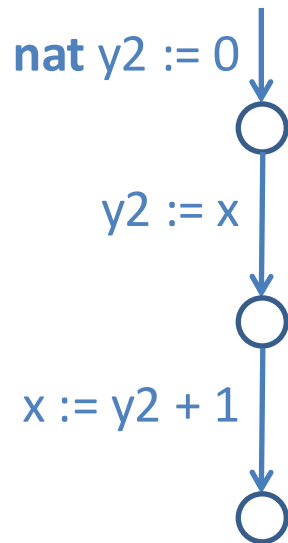
Shared Memory Programs

AtomicRegister nat $x := 0$

Process P1



Process P2



Declaration of **shared** variables
+ code for each process

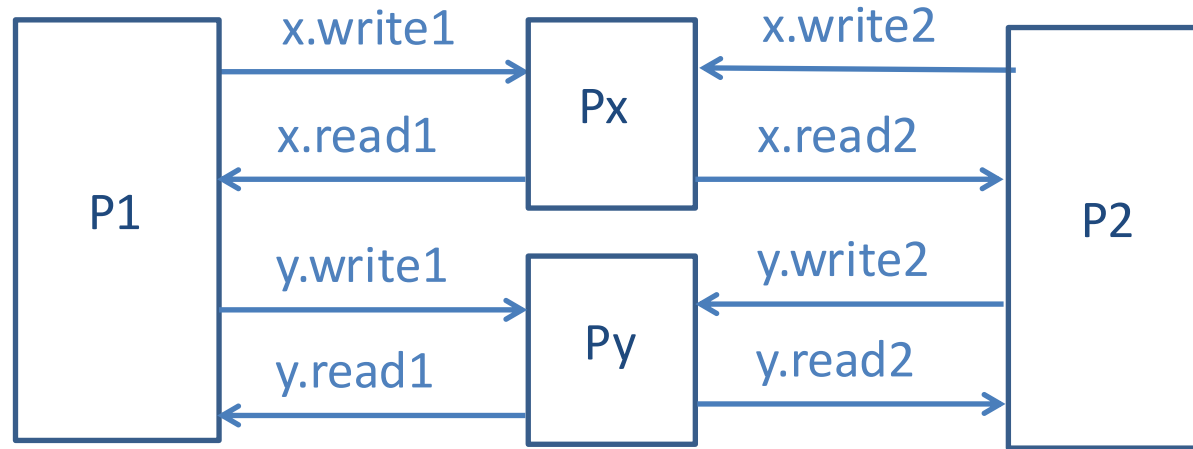
Key restriction: each process statement does one of the following

- changes local variables
- reads (single) shared var x
- writes shared var x

Execution model: execute one step of one of the processes

Can be formalized as **asynchronous processes**

Shared Memory Processes



Processes P_1 and P_2 communicate by reading/writing shared variables

Each shared variable s can be modeled as an asynchronous process P_s

For each such s

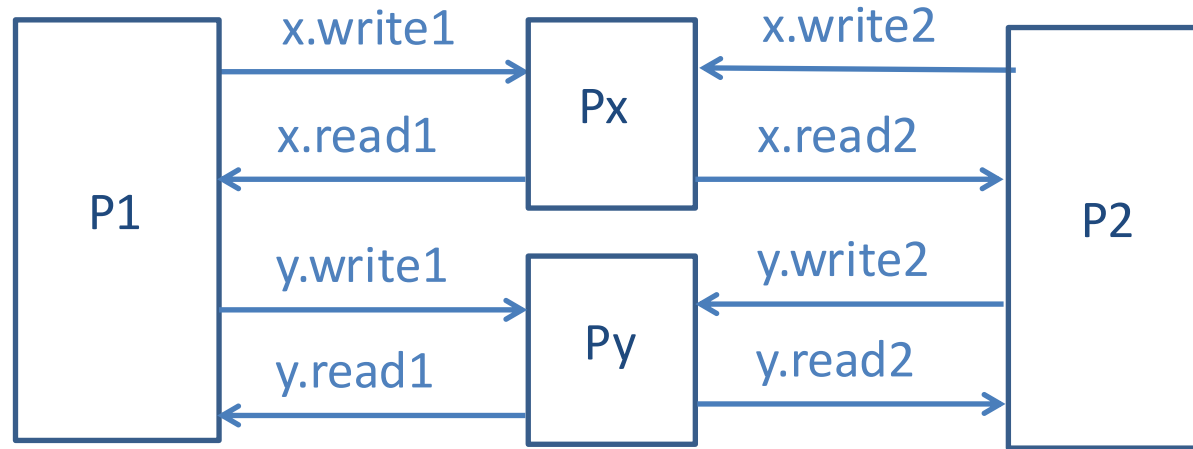
P_s has **output** channels $s.read1$, $s.read2$ and **input** channels $s.write1$, $s.write2$;

its **state** stores the value of s

Example above:

- To write x , P_1 synchronizes with P_x on $x.write1$ channel
- To read x , P_2 synchronizes with P_x on $x.read2$ channel

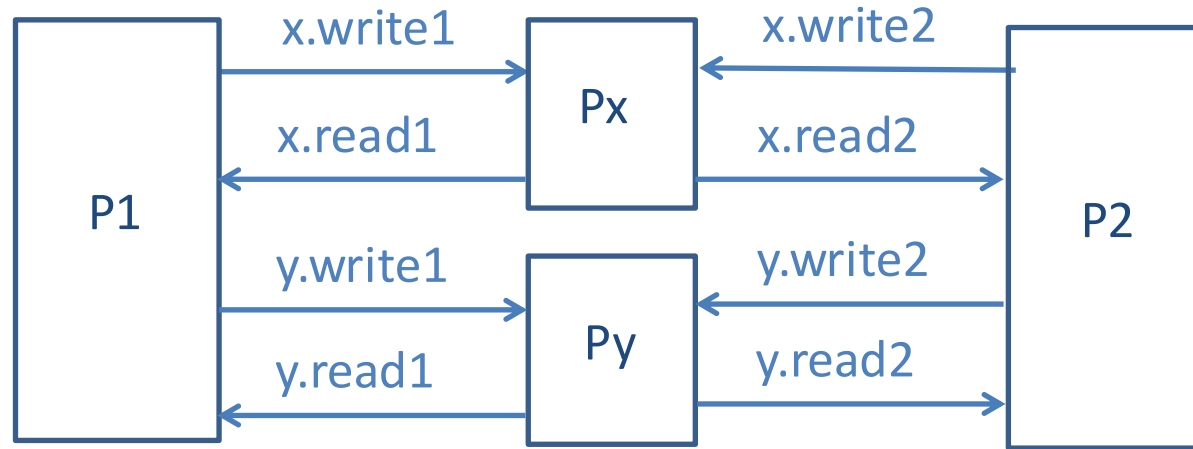
Atomic Registers



Note: By def. of asynchronous model, each step of above model is either

1. internal to **P1** or **P2**, or
2. involves exactly one synchronization
(read or write of one shared variable by one of the processes)

Atomic Registers



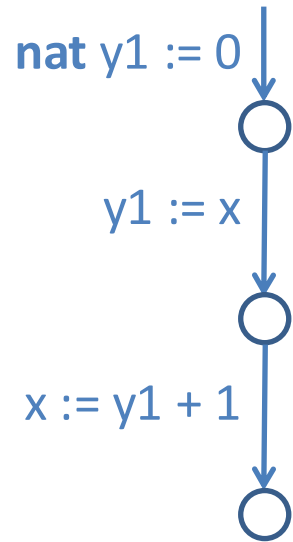
Atomic register: Basic primitives are **read** and **write**

- To **increment** such a register, a process first needs to **read** and **then write** back the incremented value
- But these **two are separate steps**, and register value can be changed in between by another process

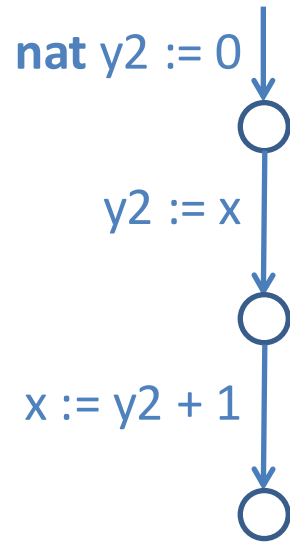
Shared Memory Programs

AtomicRegister nat x := 0

Process P1



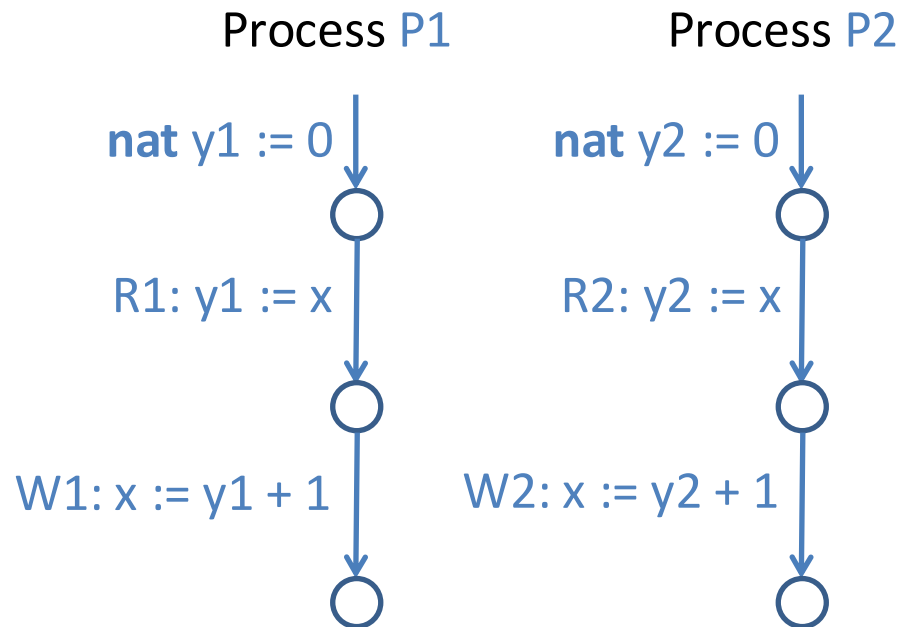
Process P2



Data Races

AtomicRegister nat x := 0

What are the possible values of **x** after all 4 read/write steps are executed?



x can be 1 or 2

Possible executions:

R1, R2, W1, W2

R1, W1, R2, W2

R1, R2, W2, W1

R2, R1, W1, W2

R2, W2, R1, W1

R2, R1, W2, W1

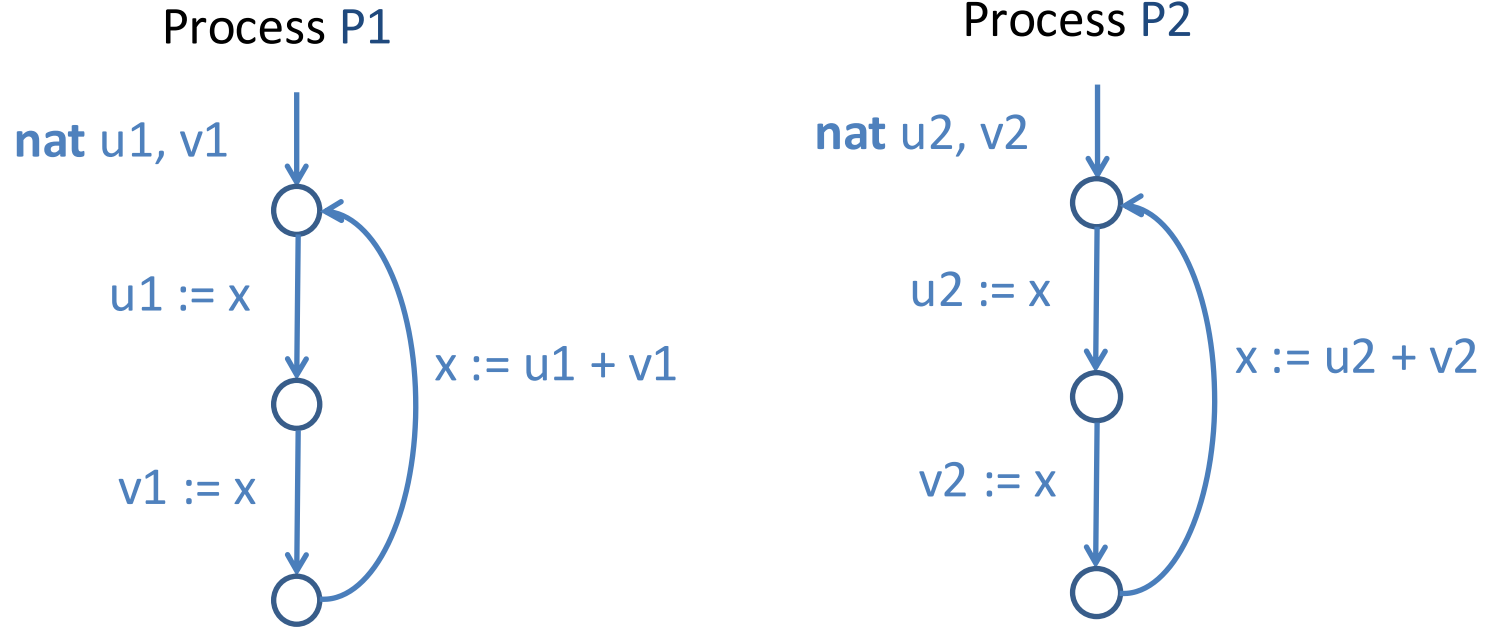
...

Data race: concurrent accesses to shared object where result depends on execution order

It should be avoided!

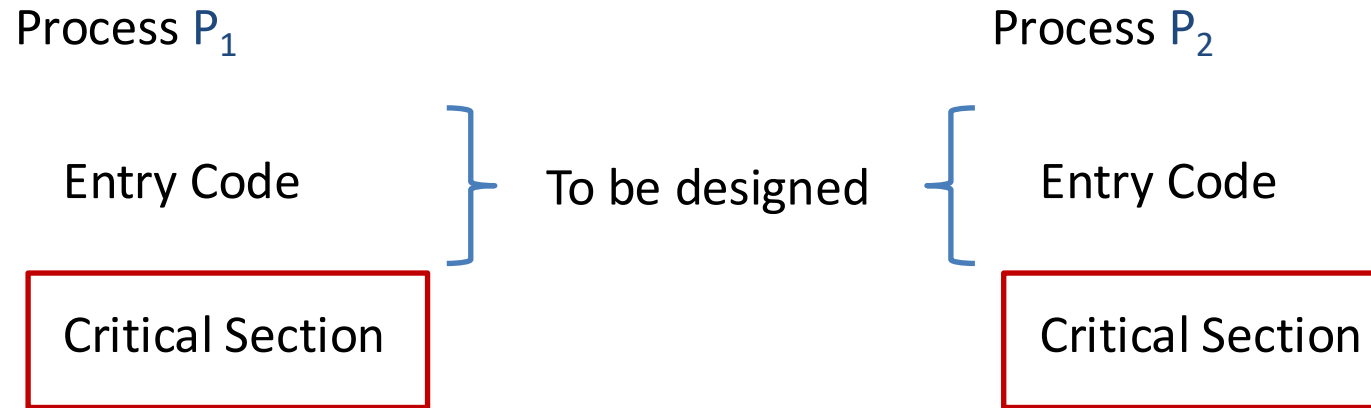
Exercise

AtomicRegister nat $x := 1$



What are the possible values for the shared register x ?

Mutual Exclusion Problem



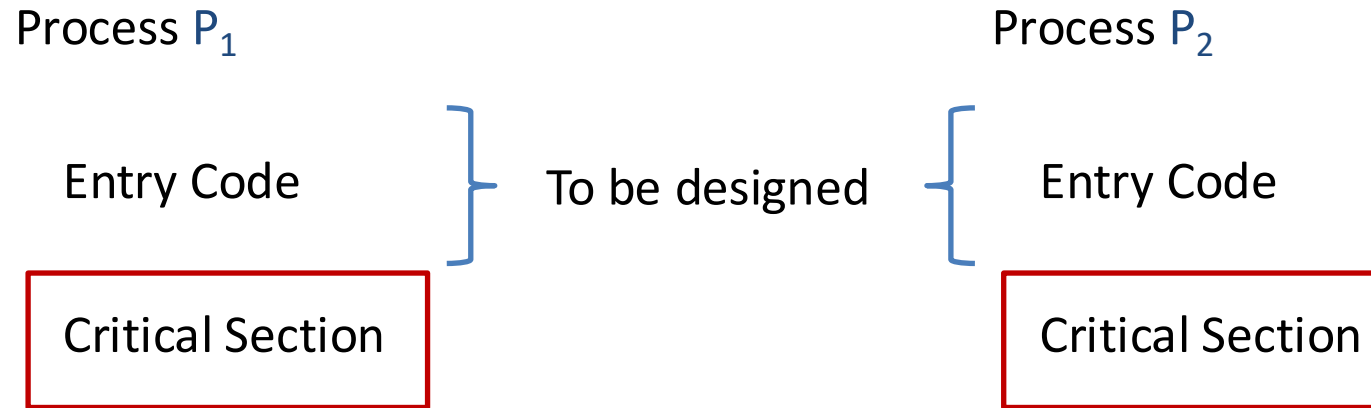
Critical Section: part of code that an asynchronous process should execute without interference from others

- Critical section can include code to update shared objects/database

Mutual Exclusion Problem: design code to be executed before entering critical section by each process

- Coordination using shared atomic registers
- No assumption about how long a process stays in critical section
- A process may want to enter critical section repeatedly

Mutual Exclusion Problem



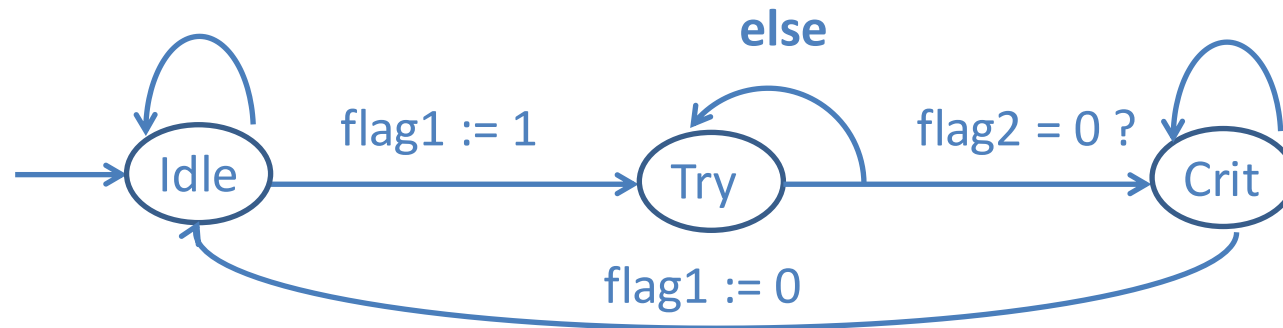
Safety requirement: no two processes should simultaneously be in their critical section
(can be formalized using invariants)

Progress requirement: if any process is trying to enter, then some process should be able to enter (no deadlocks)

Mutual Exclusion: First Attempt

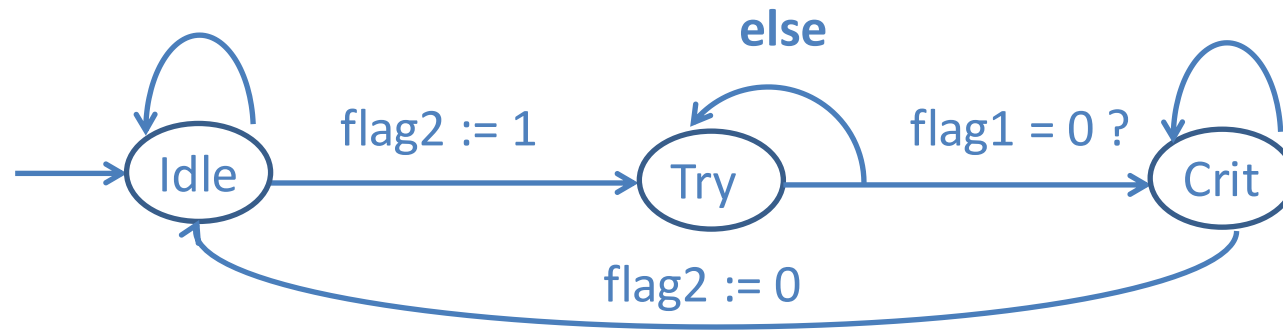
AtomicRegister bool flag1 := 0 ; flag2 := 0

Process P1



Is **safety** guaranteed?

Process P2

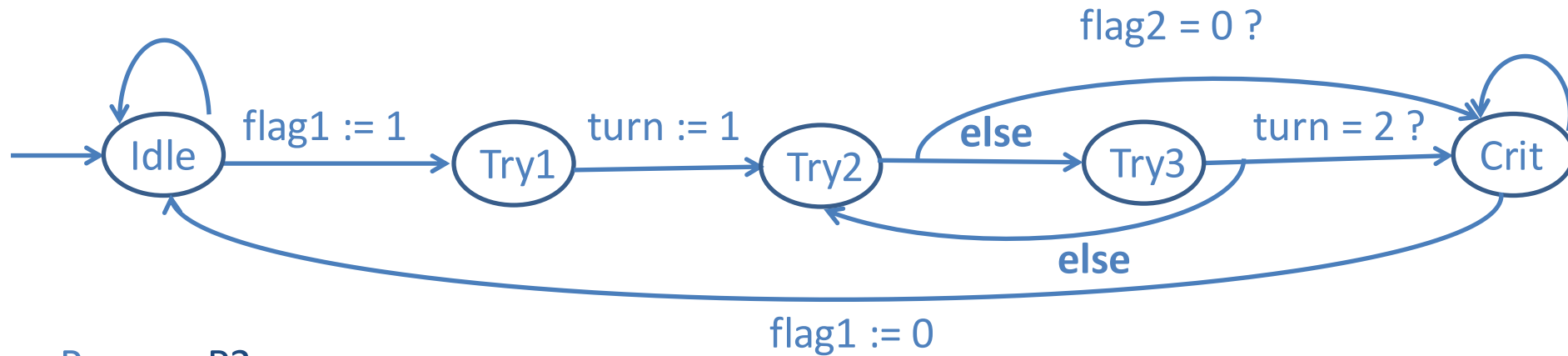


Is **progress** guaranteed?

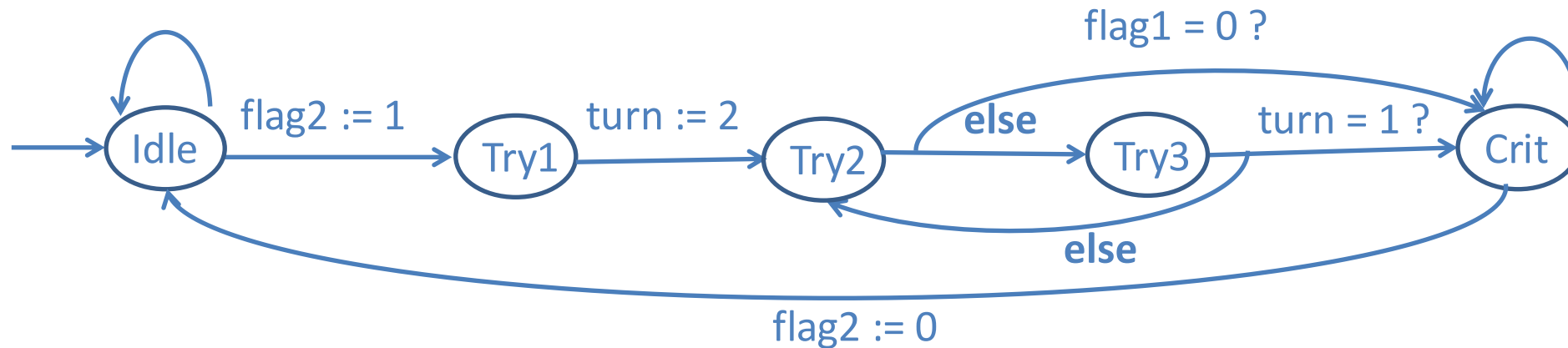
Peterson's Mutual Exclusion Protocol

AtomicRegister bool flag1 := 0 ; flag2 := 0 ; {1, 2} turn

Process P1



Process P2



Test&Set Register

Beyond atomic registers:

If we can do more than just read or write in one atomic step, then we have stronger synchronization primitives

Test&Set Register: holds a Boolean value

- *Reset operation*: changes the value to 0
- *Test&Set operation*: returns the old value and changes value to 1
- If two processes are competing to execute *Test&Set* on a register with value 0, one will get back 0 and other will get back 1

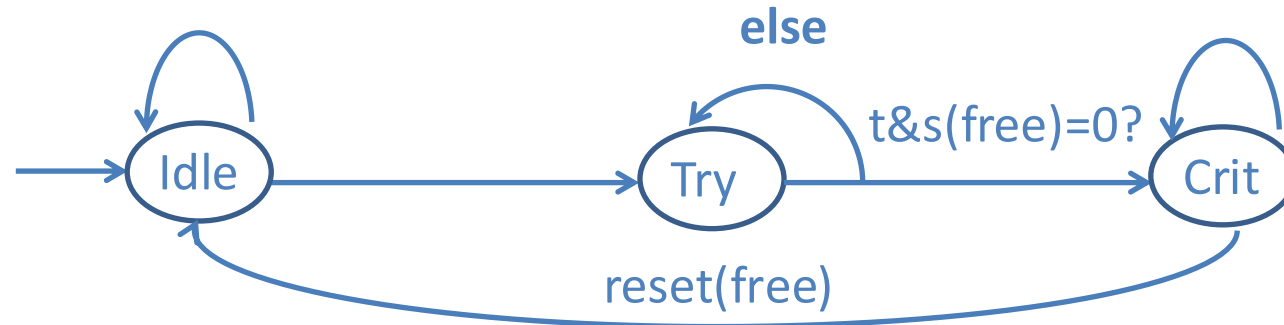
Modern processors support **strong atomic** operations

- **Example:** compare-and-swap; load-linked-store-conditional
- Implementation is more expensive than atomic read/write operations

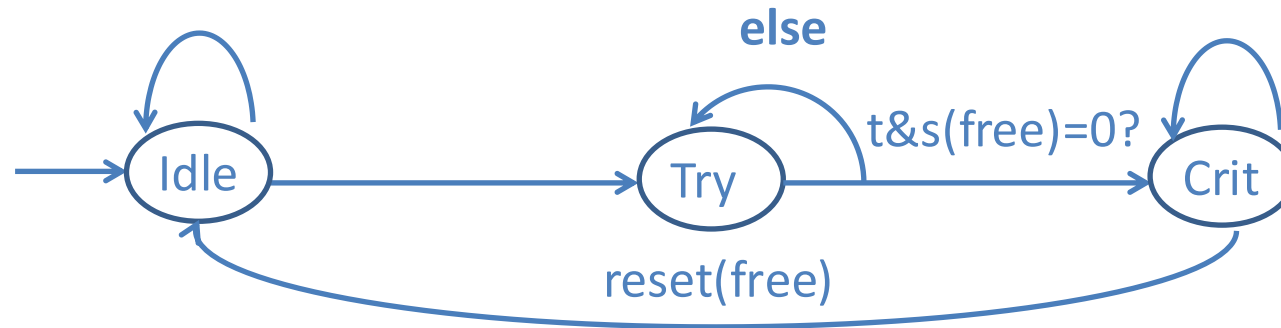
Mutual Exclusion using Test&Set Register

Test&SetRegister nat free := 0

Process P1



Process P2



Is this correct?

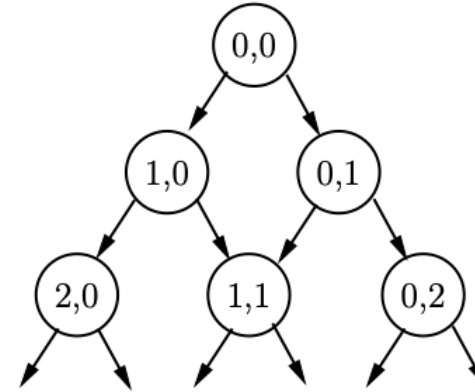
Another Look at Asynchronous Execution Model

Process P

```
nat x := 0 ; y := 0
```

```
Ax: x := x + 1
```

```
Ay: y := y + 1
```



Tasks A_x and A_y execute in an arbitrary order

Motivation: If we establish that all possible executions of this design satisfy some requirement R , then every implementation of P will satisfy R

Are the following realistic executions?

- $(0,0) \xrightarrow{A_x} (1,0) \xrightarrow{A_x} (2,0) \xrightarrow{A_x} (3,0) \dots \xrightarrow{A_x} (95,0) \xrightarrow{A_x} \dots$
- $(0,0) \xrightarrow{A_x} (1,0) \xrightarrow{A_x} (2,0) \xrightarrow{A_y} (2,1) \xrightarrow{A_y} (2,2) \dots \xrightarrow{A_y} (2,95) \dots$

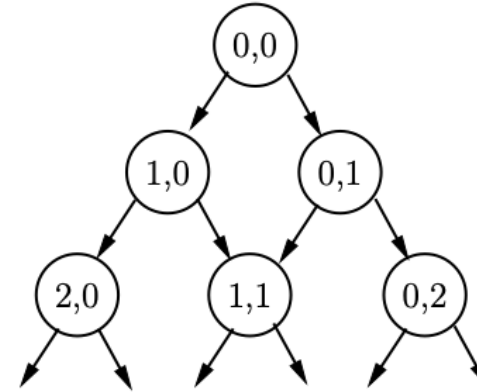
Does the system satisfy the following requirement:

In every execution, values of both x and y eventually exceed 10

Fairness Assumption

Process P

<code>nat x := 0 ; y := 0</code>
<code>A_x: x := x + 1</code>
<code>A_y: y := y + 1</code>



Fairness assumption for a task

- Assumption about the underlying platform/scheduler
- Informally, an infinite execution is *unfair* to a task if the task does not get a chance to execute

Unfair to A_y : $(0,0) \xrightarrow{A_x} (1,0) \xrightarrow{A_x} (2,0) \xrightarrow{A_x} (3,0) \dots \xrightarrow{A_x} (95,0) \dots$

Unfair to A_x : $(0,0) \xrightarrow{A_x} (1,0) \xrightarrow{A_x} (2,0) \xrightarrow{A_y} (2,1) \xrightarrow{A_y} (2,2) \dots (2,95) \dots$

Fairness assumptions restrict the set of *possible* executions to *realistic* ones *without* imposing *concrete bounds* on relative speeds

Formalizing Fairness

Process P1

<code>nat x := 0 ; y := 0</code>
<code>A_x: x := x + 1</code>
<code>A_y: y := y + 1</code>

Process P2

<code>nat x := 0 ; y := 0</code>
<code>B_x: x := x + 1</code>
<code>B_y: x = 0 → y := y + 1</code>

Definition 0. An infinite execution is *fair* to a task *A*, if the task *A* is executed repeatedly (i.e., infinitely often) during that execution

- Is this execution fair to task *A_y*? How about *B_y*?
 $(0,0) \rightarrow_{B_x} (1,0) \rightarrow_{B_x} (2,0) \rightarrow_{B_x} (3,0) \rightarrow_{B_x} \dots (105,0) \rightarrow_{B_x} \dots$
- After first step, the task *B_y* is not enabled, and so cannot be executed
This execution should not be considered unfair

Definition 1 (Weak fairness). An infinite execution is *fair* to a task *A* if, repeatedly, either *A* is executed or is disabled (*If enabled then eventually executed or disabled*)

Weak vs Strong Fairness

Process P3

```
nat x := 0 ; y := 0
```

```
Ax : x := x + 1
```

```
Ay : even(x) → y := y + 1
```

- Is this execution fair to task A_y?

$(0,0) \xrightarrow{A_x} (1,0) \xrightarrow{A_x} (2,0) \xrightarrow{A_x} (3,0) \dots \xrightarrow{A_x} (105,0) \xrightarrow{A_x} \dots$

- According to weak fairness, yes, because A_y is disabled infinitely often

Definition 2 (Strong fairness). An infinite execution is *fair* to a task A, if task A is either executed repeatedly or disabled continuously from a certain step onwards
(If repeatedly enabled then repeatedly executed)

- Above execution is weakly fair to task A_y, but not strongly fair

Fairness Assumption

Fairness assumptions for an asynchronous process **P**:

For each output and internal task, either

- no assumption,
- weak fairness assumption, or
- strong fairness assumption

Restrict the set of possible infinite executions

- If weak/strong fairness is assumed for a task **A**, then task scheduling should be such that executions are weakly/strong fair to **A**

Make it easier for a process **P** to meet a requirement **R**

- Maybe not all executions satisfy **R**, but all fair executions do

Note: In practice, fairness assumptions are environmental requirements

Requirements under Fairness Assumptions

Process P1

<code>nat x := 0 ; y := 0</code>
<code>A_x: x := x + 1</code>
<code>A_y: y := y + 1</code>

Process P3

<code>nat x := 0 ; y := 0</code>
<code>B_x: x := x + 1</code>
<code>B_y: even(x) → y := y + 1</code>

Under what fairness assumptions do **P1** and **P3** satisfy these requirements?

R1: eventually, $x + y > 10$

- **P1** and **P3** both satisfy this, without any fairness assumption

R2: eventually, $x > 10$

- **P1** with weak fairness for A_x , **P3** with weak fairness for B_x

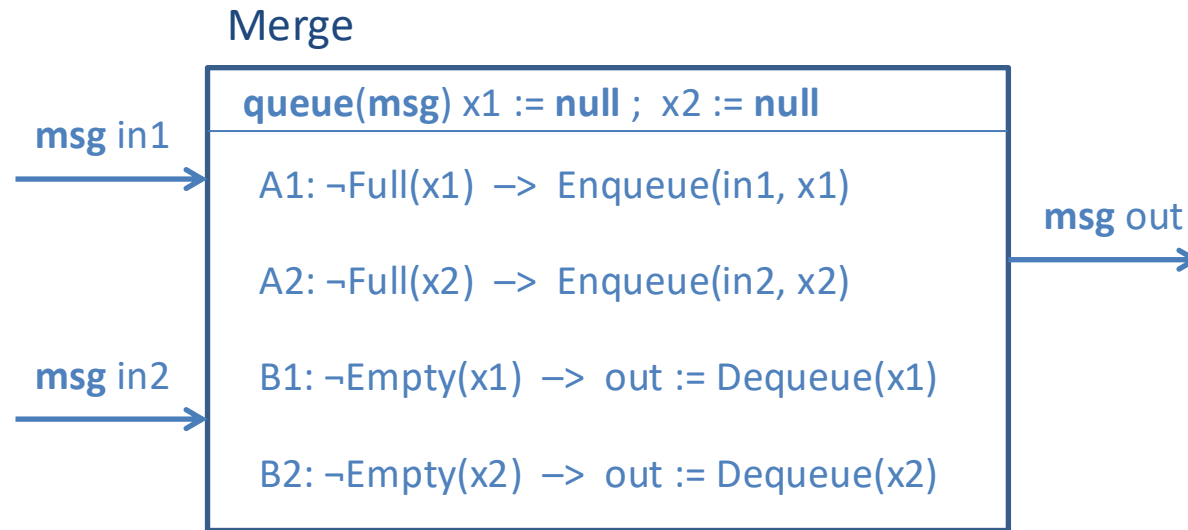
R3: eventually, $y > 10$

- **P1** with weak fairness for A_y , **P3** with strong fairness for B_y

R4: eventually, $x > y$

- neither, no matter what fairness assumption we make!

Asynchronous Merge

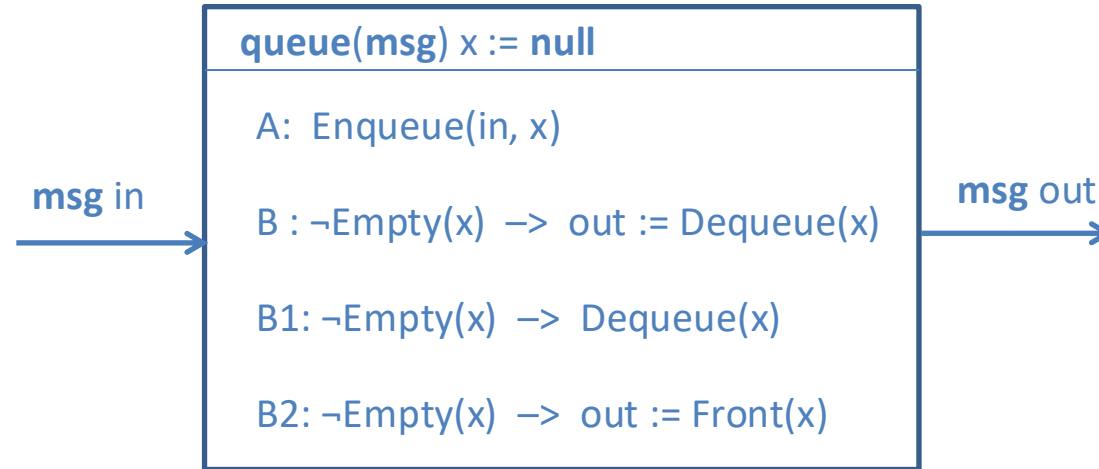


Requirement: whenever an input **message** is **received**, it is **eventually sent**

Under which fairness assumptions does the requirement hold?

Weak fairness for tasks **B1** and **B2** suffices

Unreliable (Unbounded) FIFO



Tasks A and B model normal input/output behavior

Task B1 models message loss

Task B2 models message duplication

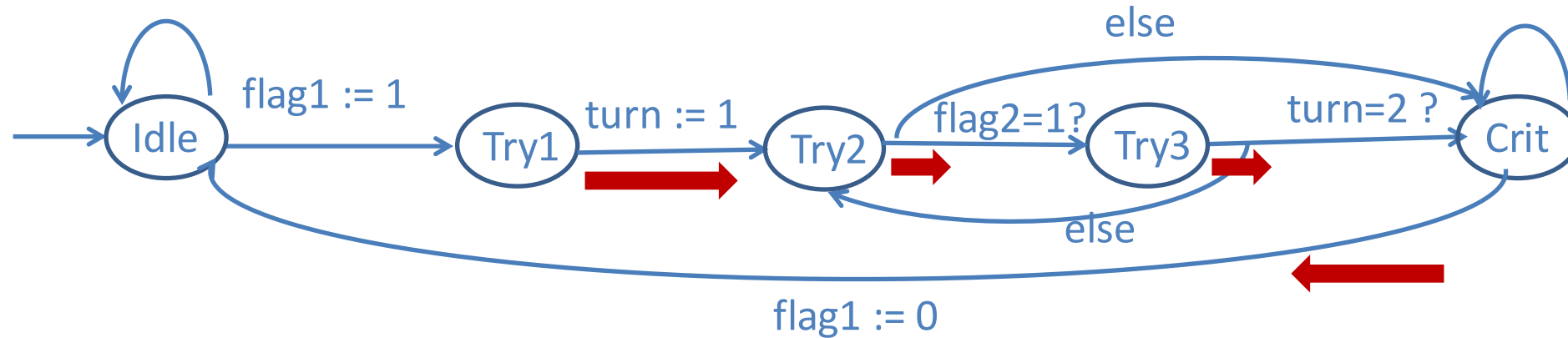
What are **natural** fairness assumptions for these tasks?

Strong fairness for B, no assumptions for B1 and B2

Fairness Assumptions for Mutual Exclusion Protocol

AtomicRegister bool flag1 := 0 ; flag2 := 0 ; { 1, 2 } turn

Process P1



Requirement: if a process ever wants to enter critical section, it eventually will

What fairness assumptions should we make?

Weak fairness for highlighted steps/tasks

Fairness Summary

A fairness assumption is an **assumption** made **about** the underlying **platform** or **scheduler**

- The weaker the assumption, the better

It **restricts** the **set of** possible infinite **executions**, allowing the **satisfaction** of **more requirements**

- Does not affect the set of reachable states and safety properties
- Does not change underlying coordination

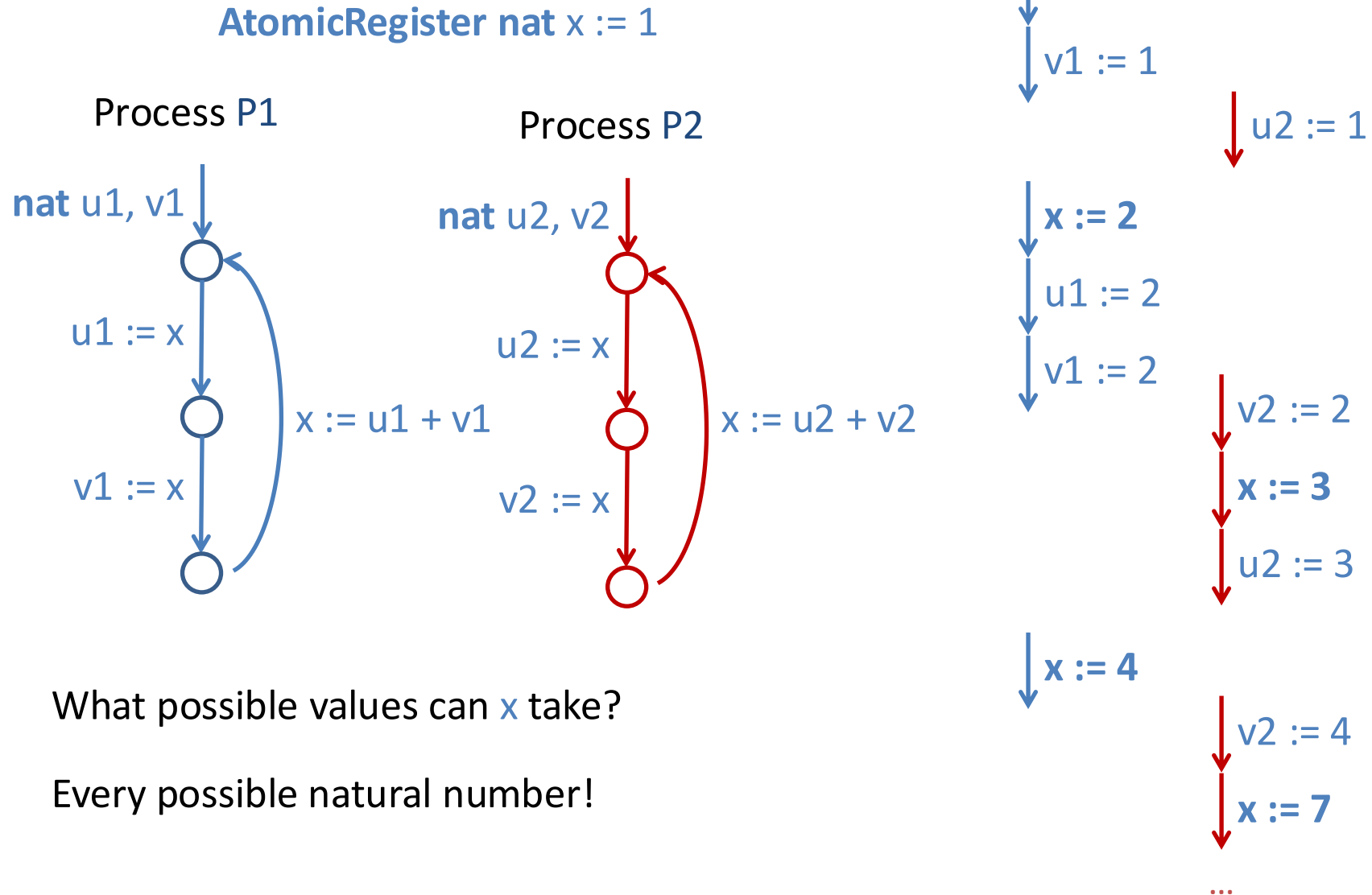
For each output and internal task, we can **assume weak** or **strong fairness**, as needed

- Strong fairness is needed if the task can switch between enabled and disabled due to the execution of other tasks

Key distinction:

- Fairness assumption for tasks (which ensures tasks get executed as expected) vs.
- *Fairness* requirements for protocols (which are about high-level goals of the problem being solved)

Exercise: Solution



Credits

Notes based on Chapter 4 of

Principles of Cyber-Physical Systems

by Rajeev Alur

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